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ECEN 5224

Week 3 Lab Report

01/28/26

*Lab 1***Objective:**

The objective of this lab is determine and analyze the difference between a signal's time-domain and frequency domain spectrums. ADS was used with a transient simulation for an ideal square wave, two square waves with a 7% and 20% period, and a 2 GBPs PRBS signal.

Method:

ADS was used to simulate the created circuit. Three ideal square waves were generated. One with an ideal rise and fall time, one with a 7% period, and another with a 20% period. In addition, a 2 GBPs PRBS signal is also generated with 100 ps rise and fall times. Each waveform had a 1V pk-to-pk amplitude run on a simulation for 200 ns at a step size of 5 ps. The frequency domain spectrum was achieved used the MeasEqn utilizing the fs() function over a frequency range of 1 MHz to 10GHz-20GHz depending. Each has a load resistor 50 Ω .

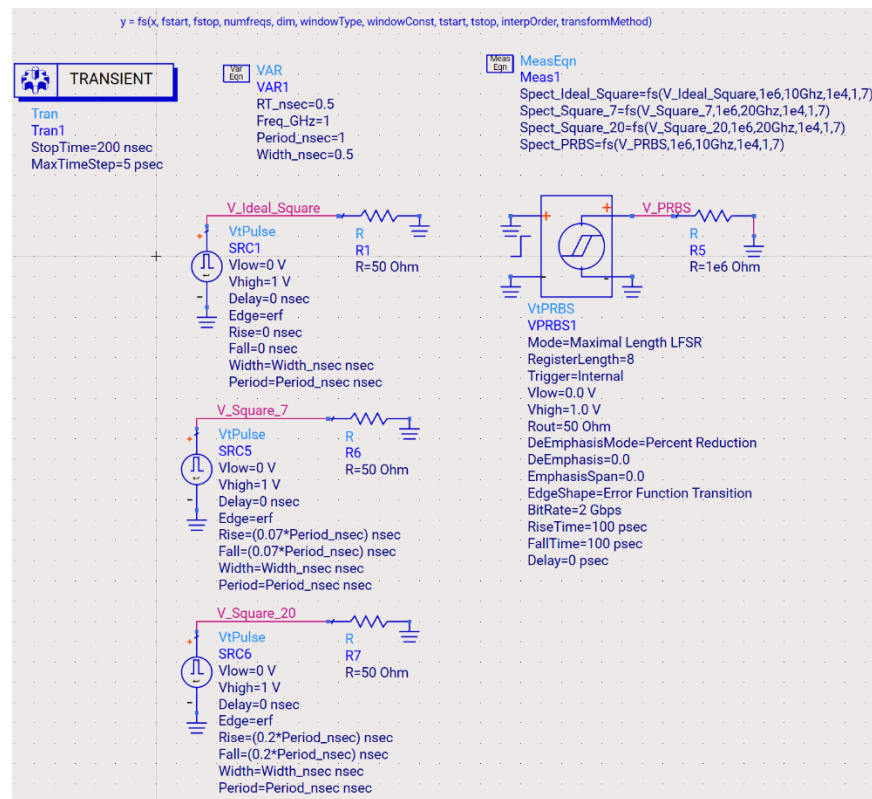
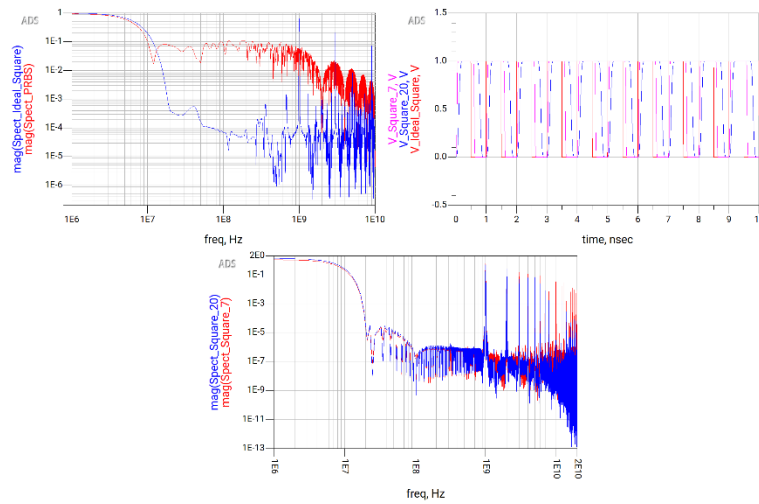


Figure 1: Lab 1 Completed Circuit

Results and Plots:



Figures 2-4: Spectrums of each waveform and the Voltage/Time plots of the three square waves

Analysis and Conclusion:

Figures 2 and 4 show the magnitude spectra of the ideal square wave, the PRBS signal, the 7% period square wave, and the 20% period square wave. The ideal square wave shows standard harmonic behavior going well into higher frequencies. A wave with zero rise and fall time like this would require a large amount of bandwidth for this to be achieved. This is different compared to the 7% and 20% period square waves. I had to extend the frequency range to 20 GHz to get all the necessary information from these two curves. The harmonics are much larger as the frequency gets larger. This is expected behavior as the relationship between bandwidth and rise time is known as $BW = \frac{0.35}{RT}$.

The PRBS spectrum appears in figure 2 alongside the ideal square wave. Although there are still oscillations, it appears to be more constant throughout the frequency spectrum. This varies from the ideal square wave as it doesn't peak as often.

This first lab shows that bandwidth is heavily influenced by rise and fall time of each waveform. Ideal signals, like the first square wave, can only exist in simulation as it is impossible to achieve a zero rise and fall time. In the real world, there isn't infinite bandwidth either and impacts from resistance, inductance, or capacitance will further change a signal's spectrum and effectiveness.

Lab 2

Objective:

The objective of this lab is take a provided IBIS model and compare its accuracy against its Thevenin equivalent under multiple different loads.

Method:

A transient simulation is run in ADS. Resistor loads of 30 Ω , 50 Ω , 100 Ω , and 10 k Ω were tested. The Thevenin resistance is tuned for each and its change in behavior is analyzed as the load increases. The input capacitance is measured for the IBIS model and compared to its C_comp value to determine what might make a difference. The package is added to the IBIS model and change in input capacitance is measured.

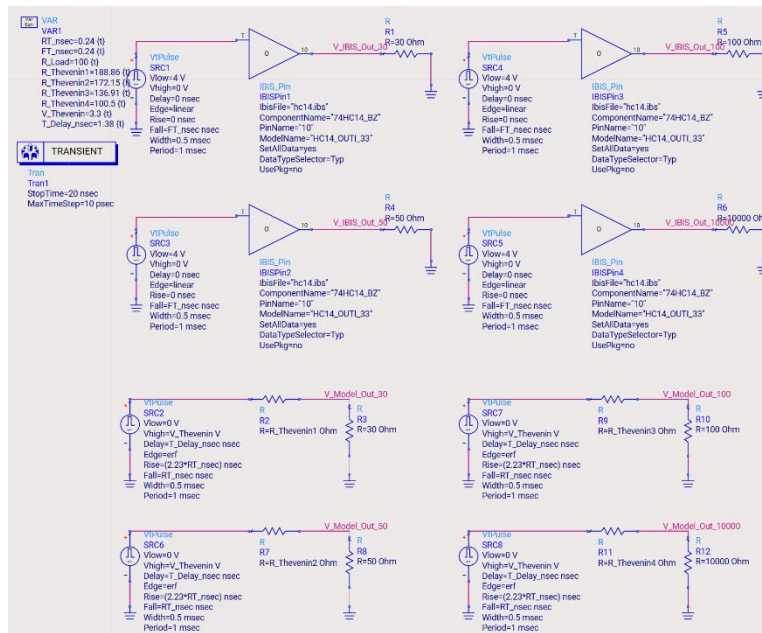


Figure 5: Lab 2 Completed Circuit

Results and Plots:

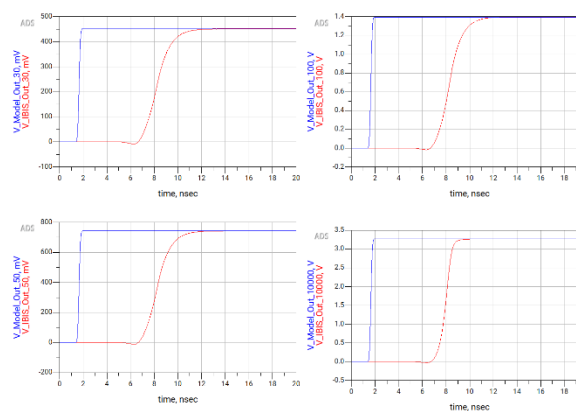


Figure 6: Transient Simulations

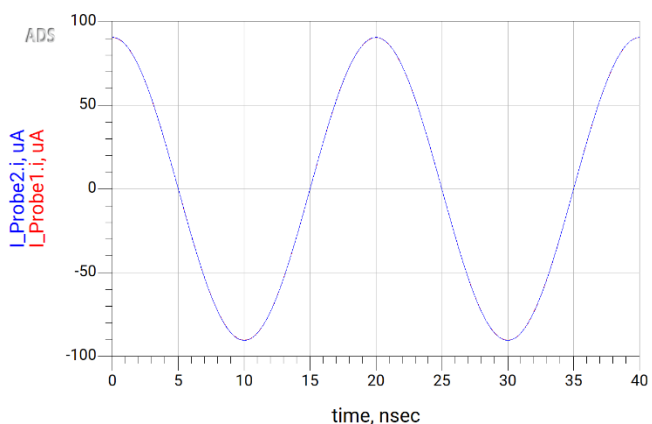


Figure 7: IBIS Input Capacitance

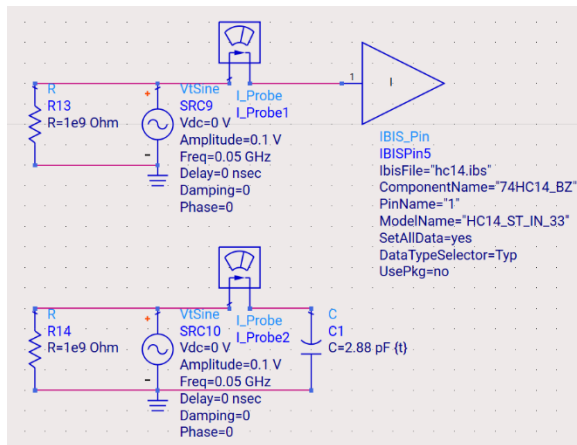


Figure 8: Measured Capacitance Circuit

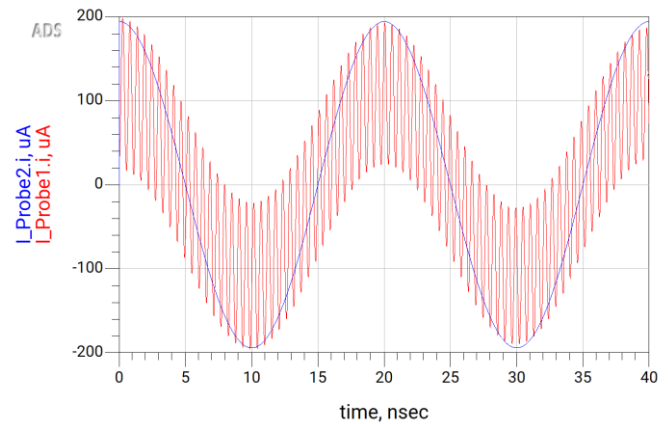


Figure 9: IBIS (with package) Input Capacitance

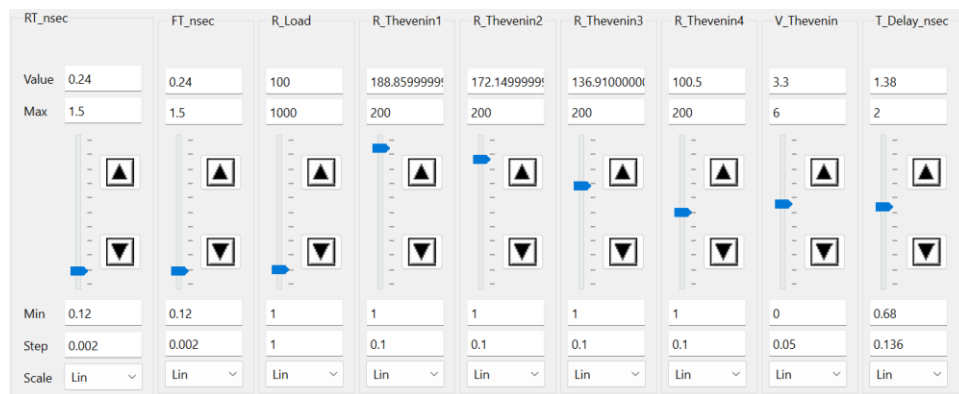


Figure 10: Tuned Values for Thevenin Resistance

Analysis and Conclusion:

The transient simulation run for the IBIS model and its Thevenin counterparts reveal a few things. It is seen that with a tune Thevenin resistance for each load, the final voltage can match the IBIS model. With each different load, the Thevenin model rises far earlier and much faster compared to the IBIS model. This shows that the Thevenin model is not the most ideal for describing the IBIS model. Under final and static steady state conditions, it works fine, but for edge shape/time the IBIS wave shows a far different behavior. The Thevenin model can be improved to further match the IBIS model but the delay would also need to be tuned.

In figure 10, the tuned values are shown which reveals the change in behavior of Thevenin resistance for different loads. The larger the resistive load, the less the Thevenin resistance is. This is what is expected from the IBIS model as it should have a nonlinear output. This means its output depends heavily on its load current.

The input capacitance of the IBIS model (without the package) is measured to be 2.88 pF from a tuned capacitor. This strongly lines up with the C_comp value of 2.8774e-12 (2.8774 pF) meaning that C_comp is a reputable parameter for the model's input capacitance. With the package added, the measured capacitance is about 6.1799 pF. This has increased the input capacitance. This is likely due to the package adding an unknown inductance, resistance, or capacitance causing the measured value to deviate from the given C_comp.